DO NOW — RECALL

# Logic circuits, adders and flip-flops

Name: _____ Class: _____ Date: _____

**1. Why can a group of four 1s remove two variables?**

**2. Simplify ¬(A ∧ B).**

**3. Calculate 1+1 in binary.**

**4. What is (1 XOR 0) AND 1?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Two variables change within the group, so only constant variables remain.

### 2. Model answer

¬A ∨ ¬B.

### 3. Model answer

$10_2$.

### 4. Model answer

1. XOR gives 1, and 1 AND 1 is 1.

### Vocabulary — know these terms

half adder — Combinational circuit that adds two one-bit inputs.

full adder — Adds two bits and a carry-in bit.

Sum — Current-column result bit.

CarryOut — Bit passed to the next higher column.

D flip-flop — Clock-triggered circuit that stores one bit.

NEW LEARNING

# Read, annotate and explain

### Find the Sum bit

XOR is 1 when A and B are different; A=1, B=1 gives Sum=0.

### Find the Carry bit

AND is 1 only when both inputs are 1; here Carry=1.

### Extend to full adder and memory

A full adder also includes CarryIn. Unlike combinational logic, a D flip-flop stores a bit until the triggering clock edge.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### A half adder receives A=0, B=1. Calculate Sum and Carry.

### Predict and explain

XOR(0,1)=1, giving Sum=1.

### Check and extend

AND(0,1)=0, giving Carry=0.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

A half adder receives A=1, B=0. Give Sum and Carry.

### Question 2

For a full adder A=1, B=1 and CarryIn=0, calculate Sum and CarryOut.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A rising-edge D flip-flop has Q=0 and D=1. D changes to 0 before the next rising clock edge. What value is stored after that edge, and why?

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A D-type flip-flop always changes Q immediately when D changes.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.